



Spatio-temporal variation of the microbiome and resistome repertoire along an anthropogenically dynamic segment of the Ganges River, India

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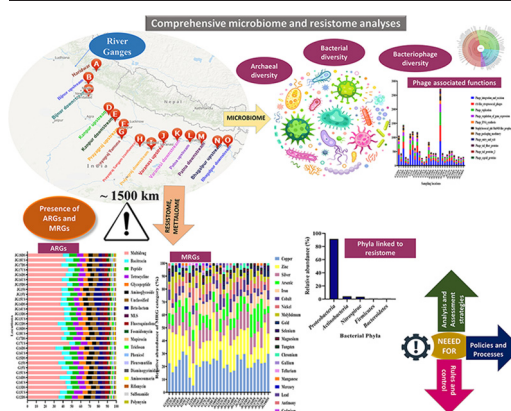
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HIGHLIGHTS

- First report of microbiome and resistome for 1500 km of River Ganges in lower stretch
- Detected resistance genes for 23 types of metals and 28 classes of antibiotics
- First report on putative host-phage linked functions metagenomic data.
- Presence of bacteriophages for pathogenic and putrefying bacteria
- Need for surveillance and mitigation strategies

GRAPHICAL ABSTRACT



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ABSTRACT

Aquatic ecosystems are regarded as a hub of antibiotic and metal resistance genes. River Ganges is a unique riverine system in India with socio-cultural and economic significance. However, it remains underexplored for its microbiome and associated resistomes along its anthropogenically impacted course. The present study utilized a nanopore sequencing approach to depict the microbial community structure in the sediments of the river Ganges harboring antibiotic and metal resistance genes (A/MRGs) in lower stretches known for anthropogenic impact. Comprehensive microbiome analyses revealed resistance genes against 23 different types of metals and 28 classes of antibiotics. The most dominant ARG category was multidrug resistance, while the most prevalent MRGs conferred resistance against copper and zinc. Seasonal differences dismally affected the microbiota of the Ganges. However, resistance genes for fosmidomycin and tetracycline varied with season ANOVA, $p < 0.05$. Interestingly, 333 and 334 ARG subtypes were observed at all the locations in pre-monsoon and post-monsoon, respectively. The taxa associated with the dominant ARGs and MRGs were *Pseudomonas* and *Burkholderia*, which are important nosocomial pathogens. A substantial phage diversity for pathogenic and putrefying bacteria at all locations attracts attention for its use to tackle the dissemination of antibiotic and metal-resistant bacteria. This study suggests the accumulation of antibiotics and metals as the driving force for the emergence of resistance genes and the affiliated bacteria trafficking them. The present metagenomic assessment highlights the need for comprehensive, long-term biological and physicochemical monitoring and mitigation strategies toward the contaminants associated with ARGs and MRGs in this nationally important river.

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1. Introduction

The occurrence of antibiotic and metal resistance genes (A/MRGs) is essentially ancient. However, various anthropogenic actions have augmented their propagation and dissemination (Yang et al., 2019). The World Health Organization (WHO) has stated ARGs as a new pollutant. Excessive metals and antibiotics in any environment impose selection pressure on the microbial communities and eventually alter the native diversity (Chen et al., 2019b). Human activities primarily influence freshwater ecosystems such as rivers and lakes and, therefore, serve as hotspots for the proliferation and evolution of ARGs and MRGs. These include several significant reservoirs that drain into the rivers such as discharges from hospitals, sewage treatment plants, industries, and agricultural/domestic/urban runoff (Li et al., 2016). The excessive accumulation of heavy metals and antibiotics in the riverine system would gradually result in bacteria trafficking ARGs and MRGs being released into the food chain, consequently menacing public health (Muurinen et al., 2022). In recent studies, the water quality deterioration of urban rivers has become an important issue, as they are pivotal in the sustainable development of cities. However, significant changes in the water quality across urban settlements have been attributed to the copious release of anthropogenic pollutants and faecal contamination, which results in the emergence of ARGs, MRGs and the associated resistant bacteria (Chen et al., 2019a; Wang et al., 2019).

The Ganges is India's national river that aids as a lifeline for millions of people. It originates from the Gangotri glacier as River Bhagirathi. It travels downward, joining the Alaknanda River at Devprayag, where its name commences as the Ganges (Khairnar, 2016a). Being an epitome of spirituality and civilization, the water of this mighty riverine system meets the diverse needs of people regarding their subsistence. The lower stretch of the Ganges is marked by its debouchment from Himalayan regions to Gangetic plains, with dense civilization along its catchment area (Bhutiani et al., 2014). Several anthropogenic activities along the basin and at the shore of this river have contributed most to its pollution. Besides, the discharge of various untreated wastes, such as industrial and sewage, has further altered the water quality of this riverine system (Jani et al., 2018). Recently, Naqvi et al. reported dumping over 50 MLD (million litres per day) of hazardous waste and 140 MLD of domestic waste in Kanpur (Haider Naqvi, 2020). Another study by Srivastava et al. mentioned that Ganges' water was unfit for bathing and drinking at Haridwar and Rishikesh (Srinivas et al., 2019).

Additionally, Ganges water at other locations such as Bihar, Kanpur, Varanasi, *Triveni Sangam* in Allahabad, Ghazipur, Buxar, Patna, Bhagalpur, and Howrah-Shivpur in West Bengal was neither potable nor suitable for bathing (Dutta et al., 2020). Faecal pollution in rivers is another significant concern, as it increases the risk of harmful microbes with antibiotic resistance. Ganges River beyond Kanpur has faecal coliforms at alarming levels, rendering it unacceptable for bathing and drinking (Srinivas et al., 2019). These pollutants ultimately result in stagnant water conditions having a colossal impact on the native microbial communities of the Ganges and also the public health (Paul, 2017a).

Previous studies on the Ganges river have documented water pollution due to excessive amounts of heavy metals such as Mn, Zn, Cr, Cu, Ni, Cd, Pb and Hg (Boral et al., 2020; Paul, 2017b). Others have reported selective isolation of faecal coliforms and antibiotic-resistant bacteria (Ali et al., 2021; Vortmann et al., 2015). Only a few metagenomics-based studies have emphasized the presence of ARGs and MRGs in this riverine system (Behera et al., 2020; Kumar et al., 2021; Reddy and Dubey, 2019; Zhang et al., 2018). However, these studies are limited to specific locations of the Ganges and lack a holistic understanding of the resistome encompassing its vast geographical area independently. Besides, the water of the River Ganges is believed to have unusual non-putrefying properties attributed mainly to its unique bacteriophage diversity (Khairnar, 2016b). However, to date, only one report encompasses this significant component of the Ganges (Kumar et al., 2021). Recent developments in high-throughput sequencing technologies have empowered us with enhanced insights into the microbial community structure and its associated functions (Sanderson et al., 2018).

Therefore, this is the first study to provide comprehensive insights into the total microbiome and the associated resistome (ARGs and MRGs) of the River Ganges about 1500 km of its anthropogenically impacted course from Haridwar to Bhagalpur along two seasons, using MinION nanopore sequencing technology.

2. Methodology

2.1. Sample collection and description of the study sites

In the present study, an approximate 1500 km long transect of the River Ganges was explored to map the reservoirs of antibiotic and heavy metal resistance genes. The sediment samples (depth ~ 20 cm), from fifteen locations, at two different seasons were collected in triplicates ($n = 90$) (n denotes the number of samples) from the left bank (LB), centre (M), and right bank (RB) of the river, in December 2017 and June 2018 (Fig. 1). The physicochemical parameters such as temperature, pH, electrical conductivity (EC), and dissolved oxygen (DO) were measured in-situ using respective probes (Supplementary Table S1). The samples were collected in 250 mL of sterile polypropylene solution bottles (Hi-Media), preserved on dry ice, and transported to the laboratory for further study. Further, each triplicate sample was pooled location-wise ($n = 15 \times 2$) and sent for a detailed elemental analysis to an accredited laboratory (Bioresource Biotech Pvt. Ltd., Pune, India) (Table S2).

2.2. Extraction of the metagenomic DNA and MinION-based shotgun sequencing

Water-saturated sediment samples were collected in triplicates. Subsequently, the samples (LB, M, and RB) were pooled together location-wise. The total DNA was extracted using RNeasy PowerSoil Total RNA Isolation Kit (Qiagen) and RNeasy PowerSoil DNA Elution Kit (Qiagen) with certain modifications as mentioned in our previous study (Samson et al., 2019). The extracted DNA samples were multiplexed with 1D Native barcoding genomic DNA EXP-NBD103 and EXP-NBD 113. The library preparation was carried out using Ligation Sequencing Kit 1D (SQK-LSK108) as per the manufacturer's instructions with certain modifications (Text S1 in Supporting information) and loaded onto FLO-MIN-106D-R9 flow cells (Oxford Nanopore Technologies). The sequencing run was carried out for ~ 56 h using the MinION sequencing platform.

2.3. Data processing and analysis

The raw reads (Fast5) were base-called using Albacore (v2.3.4). Demultiplexing of the fastq reads was done with qcat software v1.1.0 (ONT) (Vasiljevic et al., 2021). Porechop tool (v0.2.1) was employed to trim the adapters from the end of the reads and the chimeric sequences from the middle of the reads. Additionally, Porechop was run along with barcode demultiplexing to minimize the risk of cross-barcode contamination. Variations in the library sizes of the datasets were normalized using the DESeq2 package (Zhu et al., 2019). Further, the rarefaction of the data was achieved using MG-RAST (Metagenomic Rapid Annotations using Subsystems Technology). The supporting information depicts the rarefaction curves for each dataset (Fig. S1 A-C). The read statistics, quality score, and read length were assessed by Nanostats in NanoPlot (De Coster et al., 2018). Further, to eliminate the biases, the resulting data distribution was evaluated using the Shapiro-Wilk normality test (p -value threshold of 0.05), followed by the confirmation using a Q-Q plot (San et al., 2020). These rarified datasets were used then used for downstream taxonomic and functional analysis.

2.4. Microbial diversity and statistical analysis

The metagenomic datasets for microbial taxonomy and functional composition were annotated using MG-RAST. An E value cut off $1 \times e^{-5}$ and 60 % of sequence identity were the default criteria used by the MG-RAST web server for the downstream analysis. Additionally, another

metagenome classifier-Kaiju v1.7.3, was used to ascertain a higher sensitivity and precision to the taxonomic reads (Menzel et al., 2016). Singletons were removed from the (Operational taxonomic Unit) OTU file. Further OTU table was rarefied to equal sampling depth. Alpha diversity was computed using Observed OTUs (number of different OTUs in the sample), Shannon diversity index (evenness of the sample), and Simpson's index (richness and evenness of the sample) was estimated using phyloseq package in R (McMurdie and Holmes, 2013). Principal Component Analysis (PCoA) plot for beta diversity estimation was constructed using the Bray-Curtis dissimilarity matrix. Significance in the dissimilarity measure was evaluated using ANOSIM and PERMANOVA (Adonis test $p < 0.05$).

Additionally, to estimate the compositional homogeneity between the groups Betadisper test ($p < 0.05$) was applied to the datasets. To control the errors arising from multiple comparisons, Benjamini-Hochberg (BH.) FDR procedure and Bonferroni correction were used. Furthermore, to understand the influence of physicochemical and heavy metals on microbial diversity, non-metric multidimensional scaling (NMDS) plot based on the Bray-Curtis dissimilarity matrix was constructed in R using the vegan package. The envfit function in R was used to fit vectors of the physicochemical variables onto the NMDS ordination. For taxonomic analysis, ANOVA with BH correction was used to ascertain significant differences ($p < 0.05$).

2.5. Resistome analysis

The fastq files were uploaded to NanoARG to gain deeper insights into the detection and quantification of Antibiotic resistance genes (ARGs), metal resistance genes (MRG) and mobile genetic elements (MGE) (Arango-Argoty et al., 2019). The default parameter cutoff by NanoARG is (-iden 25 -cov 40 -evalue 1e-5 -prob 0.5). MRG annotation (-evalue 1e-5 -id 25 -coverage 1e-5) was carried out using the Bacmet database by NanoARG. For MGE annotation in NanoARG, the nanopore reads are aligned to a non-redundant database of the National Center for Biotechnology Information (NCBI). Kruskal-Wallis (KW) test ($p = 0.05$) was applied to the samples grouped by pre- and post-monsoon to ascertain significant ARGs and subtypes. The Centrifuge database in NanoARG was used for assigning the taxa with a particular ARG/MRG/MGE. Briefly, for each sampling location, their respective data table, post-analysis was downloaded from NanoARG. The data were sorted based on the gene name, category (ARG/MRG/MGE) and the cognate taxonomic annotation. The relative percentage of host for the respective gene category of the resistome was then plotted using GraphPad Prism v9.4.

3. Results and discussion

3.1. Resistome of the Ganges

3.1.1. Abundance, occurrence and diversity of MRGs

Various anthropogenic activities, especially from the industrial and agricultural sectors, have significantly contributed toward the release of heavy metals in large quantities, thereby impacting both soil and water. Heavy metals, unlike antibiotics, exist in the environment for a longer duration, can exert a widespread co-selection pressure, and are not easily biodegradable (Thomas et al., 2020). Likewise, several heavy metals like copper, zinc, nickel, cobalt, cadmium, mercury, and lead have antimicrobial properties. However, their persistence in an ecological niche could result in the emergence of ARGs and subsequent antibiotic resistance due to the co-selection phenomenon (Chen et al., 2019b; Pal et al., 2017). The bacteria possessing such acquired antibiotic resistance genes due to the presence of heavy metals may proliferate in the aquatic environments and can contrive into the human population, livestock, and crops in due course of time (Komijani et al., 2022). Therefore, monitoring the abundance and diversity of heavy metals in freshwater aquatic systems is necessary to assess their ecological risks. Targeting this facet, we investigated the prevalence of metal resistance genes and the associated taxa in the metagenomic datasets using the BacMet database in NanoARG. Analysis revealed the presence of MRGs against 23 various metals across all the sampling locations. The

predominant MRGs conferred resistance against copper (22.38 ± 5.97 %), zinc (21.15 ± 7.43 %), silver (11.79 ± 3.90 %), arsenic (10.09 ± 4.51 %), and iron (7.93 ± 4.13 %) (Fig. 2). However, upon seasonal comparison, no significant differences in the compositional heterogeneity of the metal resistance genes across all the locations were observed ($p > 0.05$ ANOVA with BH correction). Similar observations were reported by Reddy et al., wherein the most prevalent MRGs at the banks of the River Ganges at Varanasi were Cu, Zn, Ag, Fe, Cr, and As (Reddy and Dubey, 2019). Studies with other riverine systems, such as the Savannah River (USA) (Thomas et al., 2020), Xiangjiang River (China) (Jie et al., 2016), and Yangtze River (China) (Chen et al., 2019b) have shown comparable results with Cu, Zn, and Pb/Cd/As being the most abundant MRGs. A substantial increase of MRGs against Cu and Zn could be correlated with the contamination of river water with faecal discharges from animal farms since these trace elements are used as feed supplements, clinically used antimicrobials, and multivitamins (Chakraborty et al., 2020a). There are several reports that highlight Cu/Zn metals as the drivers for the development of antibiotic resistance in bacteria exposed to these metals. Also, wastewater influxes from various point and non-point sources and e-wastes could result in the abundance of Cu, Zn and As (Reddy and Dubey, 2019; Samson et al., 2019). Besides, the river Ganges is revered as an epitome of spirituality. It receives various metals in the form of offerings from the people, which may affect the diversity of MRGs per se. The other prominent MRGs were against cobalt, nickel, and molybdenum. Additionally, resistance genes against toxic heavy metals such as mercury (3.5 ± 0.78 %), lead (3.3 ± 0.76 %) and cadmium (1.92 ± 0.12 %) were also detected at a lower abundance in all the samples.

3.1.2. Abundance and dissemination of ARGs

Metal contaminants represent one of the chronic co-selection forces driving the antibiotic resistome in rivers. Aquatic biomes imperiled to extensive anthropogenic/urban activities are potential hotspots for the emergence of antibiotic-resistant bacteria (ARBs) (Reddy and Dubey, 2019). NanoARG-based analysis, together with MG-RAST functional profiling of the metagenomic datasets, was carried out to delineate a comprehensive resistome profile of the river Ganges. The analysis revealed the prevalence of genes extending resistance to a total of 28 classes of antibiotics. The most abundant ARGs across all the locations were those that offer resistance to multidrug/ multiple antibiotics (MDR) (39.67 ± 3.68 %), followed by bacitracin (7.70 ± 1.37 %); peptide (7.65 ± 0.95 %); tetracycline (6.86 ± 1.23 %) and glycopeptide (5.3 ± 1.14 %) (Fig. 3A). This is consistent with the studies describing a higher prevalence of MDR genes through amplicon sequencing, antibiotic susceptibility testing and isolation of MDR bacteria from the River Ganges (Ali et al., 2021; Biswas et al., 2015; Reddy and Dubey, 2019; Zhang et al., 2018). However, these studies were confined to specific locations of the Ganges, and there was a dearth of holistic insight into its resistome along the pollution ascent. To the best of our knowledge, this is the first study to comprehend a complete resistome profile of the River Ganges from Haridwar to Bhagalpur across both seasons. Besides, the presence of bacitracin, glycopeptide, and tetracycline resistance in the River Ganges suggests contamination of the water and the underlying sediment with anthropogenic inputs from various sources such as sewage and hospital/industrial/domestic waste.

Conversely, these observations vary compared to the compositional profile of ARGs diversity and dissemination of ARGs with respect to other larger rivers across the world. Studies with the Yangtze river and River Wenyu, China, showed a predominance of bacitracin resistance genes and sulfonamide resistance genes, respectively (Liu et al., 2019; Wang et al., 2019). While in the Tiber river and Aniene river, Italy, a higher prevalence of Sulfamethoxazole resistance genes was observed (Grenni et al., 2019; Patrolecco et al., 2018). Additionally, for Fales River, Canada, and Sowe River, UK, the occurrence of resistance genes toward ciprofloxacin and ampicillin was observed, respectively (Amos et al., 2014; McConnell et al., 2018). However, in River Ganges, a higher abundance of MDR genes has been observed and is consistent with the other studies, too, as mentioned above. In consideration of the expanding nature of the AMR threat, the

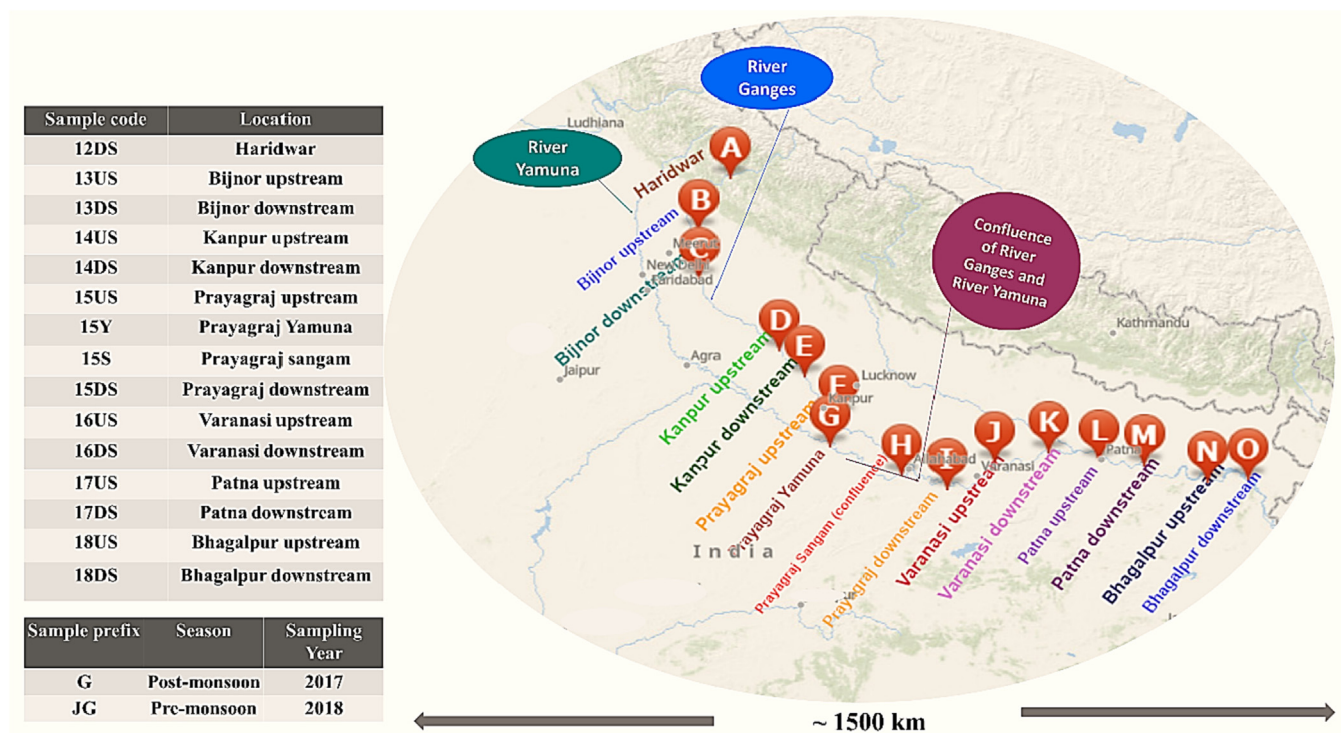


Fig. 1. Schematic illustration of the Ganges riverine system portraying the sampling locations across two seasons, pre-monsoon (2017) and post-monsoon (2018).

occurrence of MDR genes and their cognate hosts in the Ganges could pose serious threats to human health. Therefore, this study highlights the importance of the need for surveillance and integrated management measures to prevent the dissemination of ARGs and potentially resistant bacteria.

Further, to understand significant ARG categories location-wise, the upstream and downstream sampling sites were grouped into a single location (e.g., G/JG13US, DS as Bijnor). ANOVA test with a $p = 0.05$ and Bonferroni FDR corrections was applied to the grouping above. Notably, aminocoumarin ($p = 0.041$), diaminyrimidine ($p = 0.000076$), mupirocin ($p = 0.048$), and rifamycin ($p = 0.0015$) were significantly enriched ARG categories (Fig. 3B). Aminocoumarins and rifamycin are classes containing potent antimicrobial compounds produced naturally by *Streptomyces* sp. (Cheenpracha et al., 2010). In addition, mupirocin—a blend of pseudomonic acid, is produced by *Pseudomonas fluorescens* (Sutherland et al., 1985). A predominance of *Streptomyces* and *Pseudomonas* at all locations highlights that members from these genera could drive antibiotic resistance in the other native microbial community of the River Ganges. The selective pressure exerted by their antimicrobial compounds could act as a substantial driving force in shaping the microbial community structure of the River Ganges.

Further, to understand the detailed antibiotic resistance profile of the microbes in the River Ganges, categories of ARGs were investigated for their gene name (subtypes). A higher prevalence of genes *rpoB2* (13.32 ± 2.86 %), *ompR* (9.98 ± 1.83 %), *bcrA* (9.08 ± 2.05 %), *tetA* (48) (7.86 ± 1.89 %), multidrug ABC transporter (6.08 ± 1.61 %), *pata* (5.95 ± 1.33 %), *vanR* (4.98 ± 1.43 %) and *bacA* (4.57 ± 1.33 %) were abundant in all the samples as illustrated in Fig. 3C. In a signature-based-ARG source tracking study, Li et al. have linked the abundance of specific indicator ARGs to their source in each ecotype (Li et al., 2018). The occurrence of *mexE*, *mexF* and *mexT* was linked to anthropogenic stressors impacting the Kalamas and Cam Rivers. Another noteworthy observation was the presence of *tetA* and *bacA* genes in the sediment samples of the Ganges. The *bacA* gene is characteristic of the human gut and human waste, while the *tetA* gene is an essential candidate for hospital waste and wastewater-influent samples (Hu et al., 2013). Additionally, a higher incidence of *ompR*, *penA*, and *mexF* genes indicates sewage water

contamination (Gupta et al., 2019). Additionally, the presence of *bcrA*, *vanR*, *tetA*, *bacA*, *arcB*, and *kdpE*, *macB* genes signify contamination of the riverine system with human and animal faeces, respectively. These observations indicate anthropogenic stressors and the associated genes signify contamination of the Ganges riverine system with human and animal faeces, respectively.

Likewise, analysis of significant ARG groups for spatio-temporal variation revealed fosmidomycin to be more enriched in the pre-monsoon samples. At the same time, genes that confer tetracycline resistance were more prevalent in the post-monsoon season at all the locations (ANOVA, $p < 0.05$, with Bonferroni FDR corrections) (Fig. 3D). Interestingly, a total of 398 significant ARGs subtype (KW $p < 0.05$, with Bonferroni FDR corrections) were observed at all the locations in both seasons (Supporting file 2).

3.2. Microbiome of the Ganges

3.2.1. Bacterial community composition

A total of 152 bacterial phyla and 3256 bacterial genera were observed at all the locations in both seasons. *Proteobacteria* (45.89 ± 4.16 %) was the most prevalent phylum, followed by *Actinobacteria* (9.64 ± 1.46 %); *Chloroflexi* (9.36 ± 2.19 %); *Acidobacteria* (8.54 ± 3.37 %), and *Bacteroidetes* (7.35 ± 3.39 %) in all the samples across both the seasons (Fig. 4A). Besides, *Bacteroidetes* ($p = 0.0016$), *Acidobacteria* ($p = 0.0017$), *Candidatus Rokubacteria* ($p = 0.0019$), *Armatimonadetes* ($p = 0.0042$), and *Firmicutes* ($p = 0.0043$) were among the significant taxa enriched at all the sampling locations in this study (Fig. S2). Certain metagenomic studies with River Ganges have also reported *Proteobacteria* and *Actinobacteria* as the two most dominant phyla. However, these studies are limited to determining microbial diversity in the Ganges river with 16S rDNA sequences (Zhang et al., 2018) and bacterial diversity along with ARGs and MRGs in the River Ganges at Varanasi (Reddy and Dubey, 2019). Additionally, several members of *Proteobacteria* and *Bacteroidetes* phyla comprise clinically significant pathogens with multidrug resistance. Their abundance indicates contamination of water and sediments with human-associated clinical/infectious wastes (Chakraborty et al., 2020b).

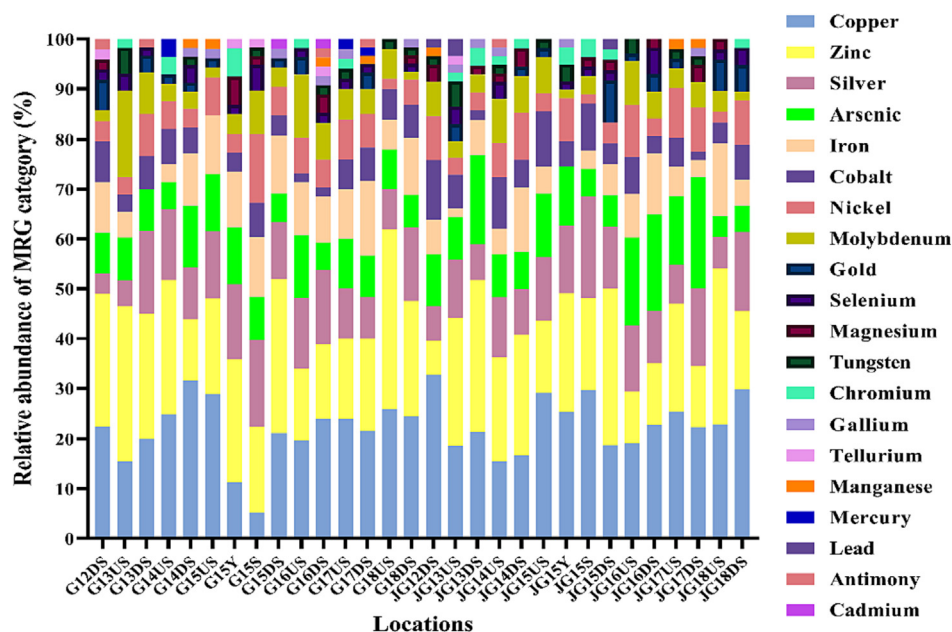


Fig. 2. Abundance and diversity of metal resistance genes (MRGs) across the lower stretch of the River Ganges.

A seasonal comparison between all the samples was made to understand compositional changes in the bacterial taxonomy. The overall bacterial composition remains similar at the phylum level across both seasons ($p > 0.99$; ANOVA with BH correction) (Fig. 4B). However, *Candidatus Handelsmanbacteria* ($p = 0.0041$), *Candidatus Kryptonita* ($p = 0.0173$), and *Candidatus Kaiserbacteria* ($p = 0.0356$) were among the significant taxa at phylum level that varied with the seasonal change. *Candidatus Handelsmanbacteria* and *Candidatus Kaiserbacteria* were significantly enriched in the post monsoon samples while *Candidatus Kryptonita* was more prevalent in pre-monsoon samples (Fig. 4C). This is the first study to report these candidate phyla in River Ganges. Besides, for other dominant phyla, although not significant yet, an increase in the relative abundance of *Acidobacteria* (6–10 %) was observed in the post-monsoon samples. In contrast, *Chloroflexi* (10 %) and *Bacteroidetes* (8 %) showed an increase in the pre-monsoon samples (dry season). This corroborates with the findings of Maestre et al., wherein the aridity affects the composition of *Chloroflexi* and *Acidobacteria*, resulting in an increase in the abundance of the former and a decrease in the latter (Maestre et al., 2015). A study on seasonal variations in the bacterial profile of Dongjiang River showed a similar pattern where phylum *Bacteroidetes* was in higher abundance during the dry season than in the wet season (Sun et al., 2017).

At the genus level, *Streptomyces* (17–32 %) dominated at all the locations except for G13US, JG14 US, J/G14DS, G15Y, JG16US, J/G16DS, where a predominance of *Nitrospira* (19–32 %) was observed. There are limited studies with freshwater systems wherein *Streptomyces* have been reported as the most abundant genera (Hamm et al., 2017; Passari et al., 2018). *Streptomyces* are considered excellent elaborators of pharmaceutical products such as antibiotics and industrial enzymes and are well known as a prominent source of novel secondary metabolites (Ma et al., 2020; Passari et al., 2018). This whole metagenome sequencing-based information could be crucial in exploring sediments of the river Ganges for the isolation of promising strains of *Streptomyces* and its potential bioprospection. Other prominent genera include *Pseudomonas* (16.08 ± 12.65 %), *Bradyrhizobium* (4.55 ± 1.55 %) and *Sphingomonas* (4.75 ± 1.43 %) (Fig. 4D). Seasonal variation did not have a significant impact on the bacterial diversity at the genus level ($p > 0.99$; ANOVA with BH correction) (Fig. 4E). However, a total of 127 significant bacterial genera varied with the season while 475 others differ spatially ($p < 0.05$) (Supporting file 2).

Pseudomonas was predominant in JG15Y, and J/G15DS ($n = 28$ –67 %). However, a significant increase (Wilcoxon test p -value = 0.0156) in the genus *Pseudomonas* was observed at G15DS (67.24 %); JG15DS

(45.97 %). This could be correlated with the substantial increase in the relative abundance of *Proteobacteria* at J/G15Y (50.4 ± 2 %) and J/G15DS (55.2 ± 5.1 %). This result corroborates our previous study where the confluence of Yamuna River with the Ganges at *Triveni Sangam*, *Prayagraj*, results into a remarkable increase in the *Pseudomonas*. This could be an attribute of the aforementioned confluence, with excessive human interference alongside the banks of the River Ganges (Samson et al., 2019). Additionally, the presence of *Pseudomonas* as the dominant genus at Bijnor could be due to the discharge of around 24 Million Liter Per Day (MLD) of wastewater (WW) (17 %) from the four-point sources of drains, namely, Hemraj Drain; Bijnor Sewage Drain; Chhoiyad drain and Malan River (carrying sewage of Nazibabad town and later merging with river Ganges at Bijnor) (CPCB (Central Pollution Control Board), 2013).

3.2.2. Archaeal community composition

River Ganges has been underexplored for archaeal diversity. Aiming for this, we analyzed our metagenomic datasets for dissecting the archaeal diversity. Analysis revealed the presence of a total of 22 archaeal phyla and 234 archaeal genera. A higher prevalence of *Euryarchaeota* was observed in all the samples with a mean relative percentage of (38.93 ± 13.54 %), followed by *Thaumarchaeota* (37.91 ± 19.97 %); *Candidatus Bathyarchaeota* (6.23 ± 3.91 %); *Candidatus Woesearchaeota* (3.14 ± 2.46 %); and *Crenarchaeota* (2 ± 0.45 %) (Fig. 5A). Although a predominance of *Euryarchaeota* was observed at the phylum level, members of phylum *Thaumarchaeota* were more enriched at the genus level. A significant fraction was contributed by *Candidatus Nitrosotenuis* (19.75 ± 12.76 %); *Nitrosarchaeum* (18.70 ± 13 %); *Candidatus Methanoperedens* (8.52 ± 9.43 %); and *Methanosarcina* (7.29 ± 5.95 %). Several studies have suggested a prevalence of AOA and methanogens in various wastewater (WW), WW treatment plants, and activated sludge (Ibarbalz et al., 2016; Płaza et al., 2021; Sharma et al., 2012; Vítězová et al., 2020). The presence of these genera in the River Ganges indicates water contamination through anthropogenic interference, which might act as a driving force toward the formation of the archaeal community structure (Fig. 5B). Further, a seasonal comparison of the samples for scoring differences in the archaeal diversity at phylum and genus level, revealed no significant differences (ANOVA with BH correction) (Fig. 5C, D). To our knowledge, this is the first study summarizing archaeal community composition along ~1500 km transects of River Ganges for two seasons.

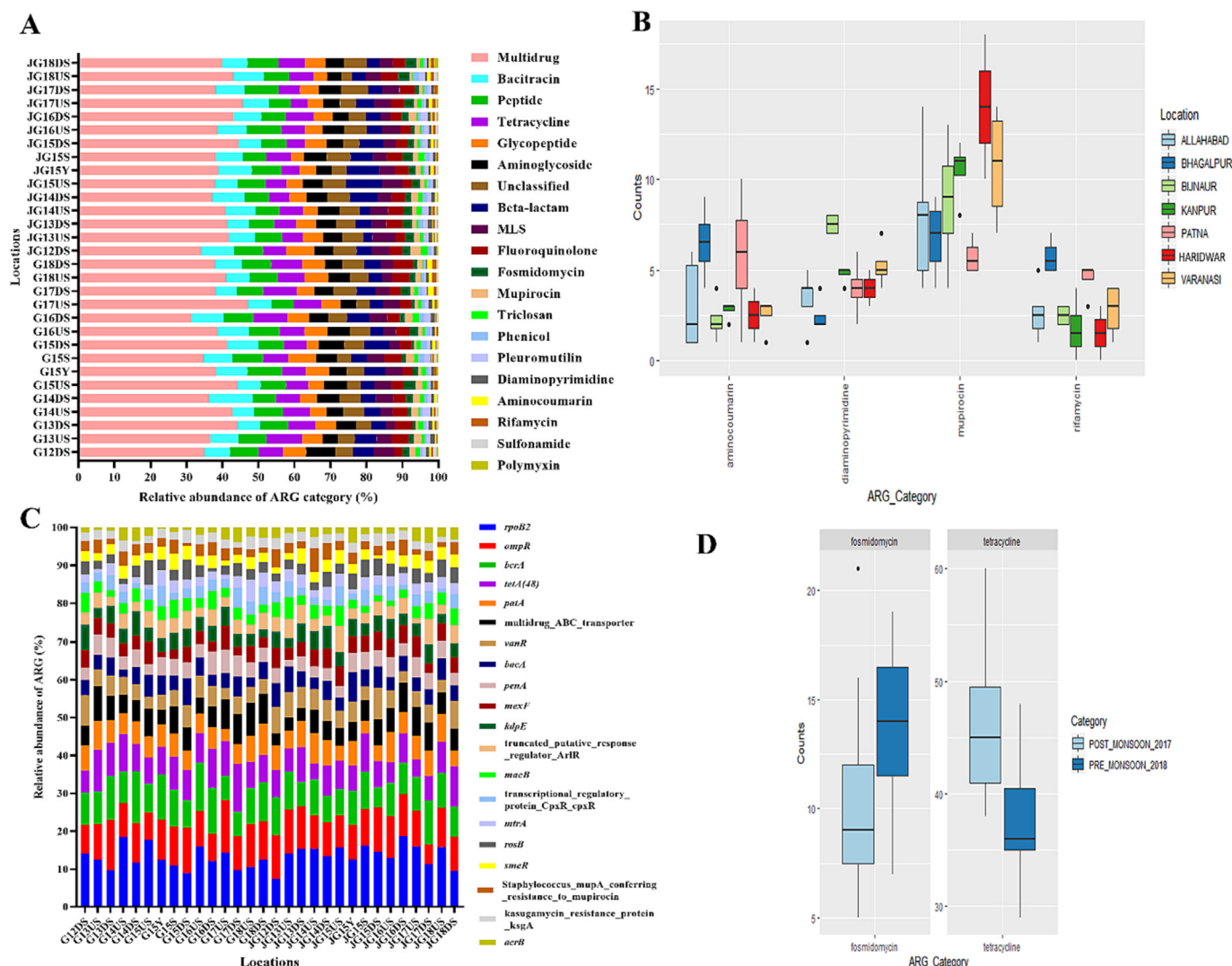


Fig. 3. Abundance and diversity of antibiotic resistome across the lower stretch of River Ganges. **3A:** Graph illustrating categories of antibiotic resistance genes. **3B:** Significantly enriched categories of antibiotic resistance genes. **3C:** Relative abundance depicting the top 20 genes encompassing the antibiotic resistome. **3D:** A seasonal comparison of the significant ARG categories.

3.2.3. Viral community composition

River Ganges harbour different microbial life yet remains unexplored for its viral diversity. In this study, we have attempted to delve into the diversity of dsDNA viruses in River Ganges from Haridwar to Bhagalpur over two seasons. At the order level, a significant fraction was assigned to unclassified viruses ($62.55 \pm 9.79\%$), followed by *Caudovirales* ($33.55 \pm 8.49\%$) and *Herpesvirales* ($3.90 \pm 8.49\%$) (Fig. 6A). As compared to the prokaryotic and eukaryotic sequences in the metagenomic datasets, the viral sequences are underrepresented in the reference databases. This, in turn, inevitably impacts the reported viral diversity in the environmental metagenomes and results in a large proportion of unclassified viral sequences (Palermo et al., 2021). The presence of several unclassified viruses in the River Ganges opens the door for exploring its untapped viral diversity for various applications. Another identifiable fraction includes the order of bacteriophages *Caudovirales*. This observation is consistent with the viral diversity of other freshwater systems, where *Caudovirales* accounted for numerical superiority (Cai et al., 2016; Ge et al., 2013; Lu et al., 2022). Further classifying the viral reads at the family level revealed a predominance of the family of giant viruses- *Mimiviridae* ($31.68 \pm 11.16\%$), followed by *Siphoviridae* ($19.09 \pm 5.06\%$), *Myoviridae* ($11.23 \pm 4.44\%$) and *Phycodnaviridae* ($7.38 \pm 2.83\%$) (Fig. 6B). Certain metagenomics-based studies on the viral diversity of eutrophic rivers and lakes have revealed

the dominance of *Mimiviridae* and bacteriophage (*Siphoviridae*, *Myoviridae* and *Podoviridae*) families (Malki et al., 2020; Nogueira et al., 2022; Palermo et al., 2019; Runa et al., 2021). Similarly, for River Ganges, a significant pollution load from various anthropogenic activities commences downstream of Haridwar (CPCB (Central Pollution Control Board), 2017). A higher prevalence of these viral families could be attributed to the resulting magnitude of water contamination, which leads to deterioration in the water quality of the river Ganges, thereby modulating the microbial community structure. Additionally, seasonal changes dismally affected the viral fraction (ANOVA $p > 0.05$ with BH correction) (Fig. 6 C, D).

3.3. Estimation of microbial diversity indices

Alpha diversity of the Archaeal community across two seasons was assessed using observed Shannon and Simpson's diversity indices. Although not significant, a higher number of OTUs were observed post-monsoon season. However, a relative increase in the Shannon diversity (5.42) and Simpson's index (0.98) was observed for the Archaeal community, indicating a higher richness and evenness for the pre-monsoon samples (Fig. S2A). Beta diversity was computed to understand the microbial compositional changes, including seasonal differences, using the Bray-Curtis dissimilarity matrix. A lower dissimilarity was observed between the samples across

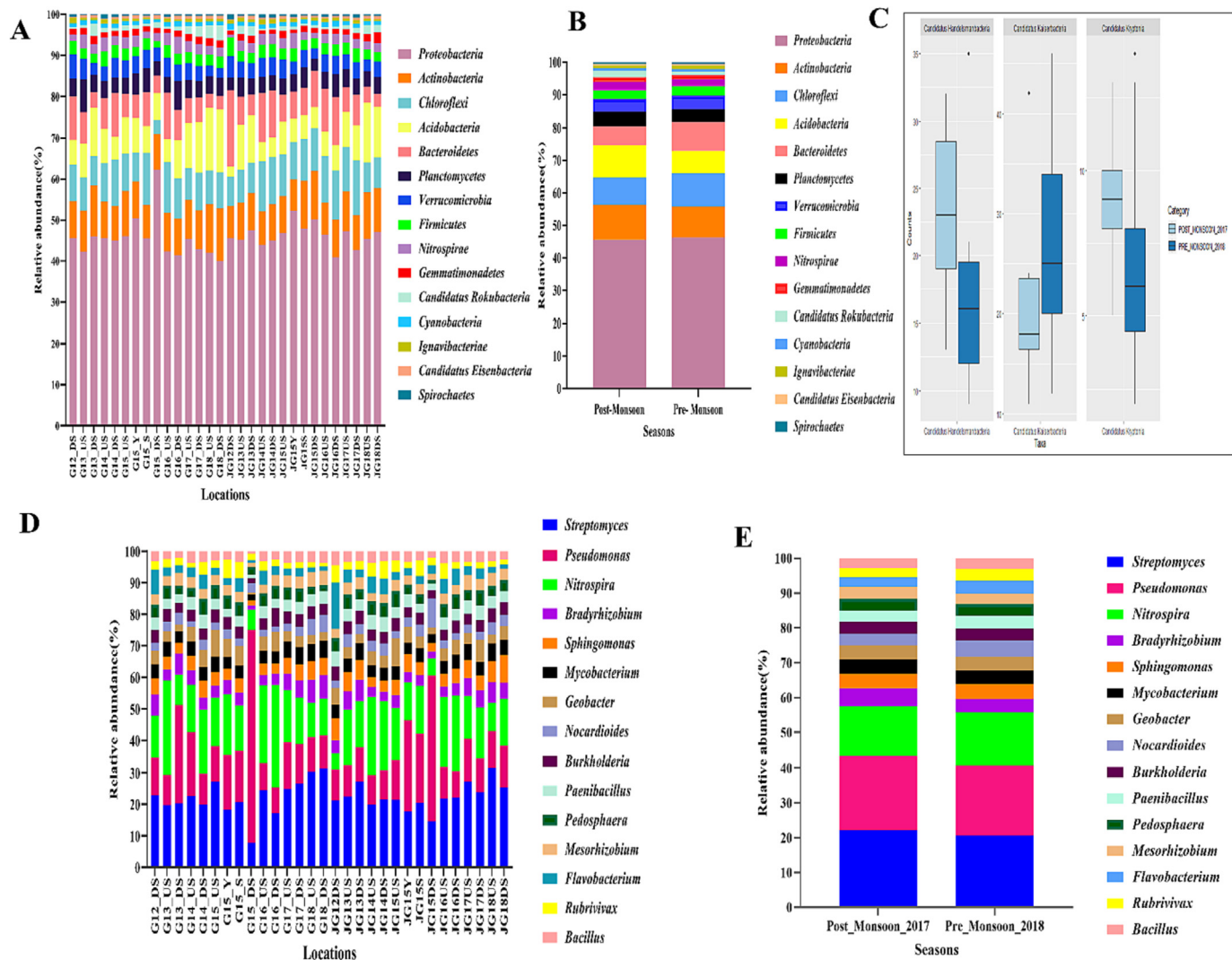


Fig. 4. Bacterial community composition across the lower stretch of River Ganges. **4A:** Relative abundance of top 15 bacterial phyla. **4B:** A seasonal comparison of the top 15 bacterial phyla showing compositional homogeneity. **4C:** Graph showing the significant taxa at the phylum level that varied with the seasonal change. **4D:** Relative abundance of top 15 bacterial genera. **4E:** A seasonal comparison of the top 15 bacterial genera showing compositional homogeneity.

both the seasons on axis 1 with 39 % variation (Adonis test $p > 0.05$), indicating no significant influence of seasonal variation on the Archaeal diversity (Fig. S2B).

3.4. Influence of physicochemical factors and heavy metals on microbial diversity

Microbial communities in a riverine system are greatly influenced by their physicochemical parameters and the extent of heavy-metal contamination (Huang et al., 2021a). In the current study, we have assessed the effect of physicochemical properties such as pH, temperature, DO, and heavy metals (As, Cd, Cr, Hg, Ni, Pb, and Zn) on the microbial composition for both seasons. A higher concentration (mg/kg) of Zn (32–57) followed by Cr (8–50), Ni (4–14), and Pb (1–12) was observed at all the locations. However, the concentration of Hg was found to be enriched only in a few of the downstream sites of G12DS, JG13US, JG15US, and G17US. Seasonal changes dismally affected the prevalence of metals in the sediments of the river Ganges ($p > 0.05$ ANOVA with BH correction) (Fig. S3). Our results are consistent with other studies on the river Ganges in which a higher abundance of Cr, Co, Ni, Cu, Cd, Zn, Mn and Pb metals through elemental analyses have been reported in its downstream (anthropogenically impacted) regions (Boral et al., 2020; Pandey and Singh, 2017). Further, we also explored the association among the physicochemical parameters, elemental variables and microbial community. Several studies have made

known that long-term exposure to heavy metals (Cu, Cr, Ni, and Zn) alters the microbial community structure of the soil to varying degrees (Bååth et al., 1998; Huang et al., 2021a; Zorz et al., 2019). The NMDS plot showed that temperature and heavy metals such as Hg, Cr, Cl, Zn, Pb, and Ni were the key drivers positively influencing the microbial diversity for all the sites across both seasons. In contrast, DO, pH, and arsenic showed a higher negative correlation with the axis than cadmium (Fig. S4).

3.5. Taxonomic diversity and functions of microbes comprising the resistome of Ganges

3.5.1. Microbial hosts and associated functions for MRGs

In addition, we analyzed our data to understand the association between the microbial communities and the MRG categories. *Proteobacteria* comprised a significant fraction (91.37 %) of members contributing to the metal resistome repertoire of the river Ganges, followed by *Actinobacteria* (4.31 %), *Nitrospira* (3.44 %), *Bacteroidetes* (0.86 %), and *Firmicutes* (0.86 %) (Fig. 7A). Subsequently, at the genus level, the taxonomic annotation revealed a total of 116 Archaeal species carrying MRGs conferring resistance to mainly Cu, Zn, As, Ag, Mg, and Hg (Table S3). The entities were clustered into higher taxonomic rankings at the genus level, and the relative abundance of genera was plotted. A predominance of *Pseudomonas* (28.44 %); *Burkholderia* (4.31 %), *Geobacter* (4.3 %); *Xanthomonas* (3.44 %),

and *Pandoraea* (3.44 %) was observed for a majority of the MRG categories (Fig. 7B).

Transition metals like iron, cobalt, nickel, copper, and zinc are prerequisite for several biological functions and virulence mechanisms in most microbes (Palmer and Skaar, 2016). To unveil the functions associated with each MRG type, we investigated our metagenomic datasets in MG-RAST (KEGG database). Analysis assigned maximum functions related to copper and zinc homeostasis. A higher relative abundance of Copper-translocating P-type ATPase (CopA) (26.77 ± 1.97 %) followed by Response regulator of zinc sigma-54-dependent two-component system (ZraR) (17.76 ± 3.32 %) and multicopper oxidase (CueO) (12.13 ± 1.91 %) was observed in all the samples (Fig. 7C). CopA and CueO are crucial for maintaining copper homeostasis amid environmental fluctuations (Rensing and Grass, 2003). Since MRGs conferring resistance against copper are more enriched in all the samples, the occurrence of the corresponding functional components is evident. Besides, certain studies have shown that resistance to copper and zinc offers the members of the genus *Pseudomonas* and *Burkholderia* virulence advantages (Schaefer, 2020; Virieux-Petit et al., 2022). These observations were consistent with the above discussion on MRGs-related hosts. The other prominent proteins include arsenical-resistance protein Acr3 (8.94 ± 2.70 %), Co-Zn-Cd resistance protein Czc (6.51 ± 1.39 %) and Mg/Co efflux protein CorC (2.95 ± 0.67 %). Members of the Acr3 family are involved in the biotransformation of arsenic in different microbes, thereby offering them a survival advantage

in metal-contaminated sites (Desoeuvre et al., 2016). A maximum abundance of functions associated with Cu, Zn, As, and multiple metal resistance suggests persistent exposure of the microbial community to the metal-contaminated urban/industrial discharges. Besides, most gram-negative and pathogenic bacteria possess multiple metal resistance genes, such as proteins from CorC and Czc families (Huang et al., 2021b; Nies, 1992). The taxa assigned to MRGs comprise gram-negative bacteria that are either pathogens or opportunistic, causing various human-associated infections. This is the first study on the river Ganges to comprehend metal resistome and the associated microbial diversity from Haridwar to Bhagalpur (~1500 km) across two seasons.

3.5.2. Microbial hosts and associated functions for ARGs

The taxonomic annotation of the datasets was carried out to analyze the association between ARGs and microbial taxa. The Archaeal domain (99.5 %) comprised a significant fraction of the ARGs. The major constituents among them include *Proteobacteria* (85.62 %), *Actinobacteria* (6.98 %), *Bacteroidetes* (1.2 %), and *Chloroflexi* (1.12 %) (Fig. S5A). At the genus level, *Pseudomonas* (35.6 %), *Burkholderia* (11.8 %), *Geobacter* (8.8 %), *Streptomyces* (5.5 %), *Stenotrophomonas* (5 %), *Xanthomonas* (4.5 %), *Ramlibacter* (4 %), *Pandoraea* (4 %), *Cupriavidus* (3.4 %), and *Ralstonia* (3.1 %) were the top 10 most abundant genera in all the locations (Fig. S5B). A similar observation was reported by Reddy et al., wherein *Pseudomonas* was found to be the dominant genus contributing

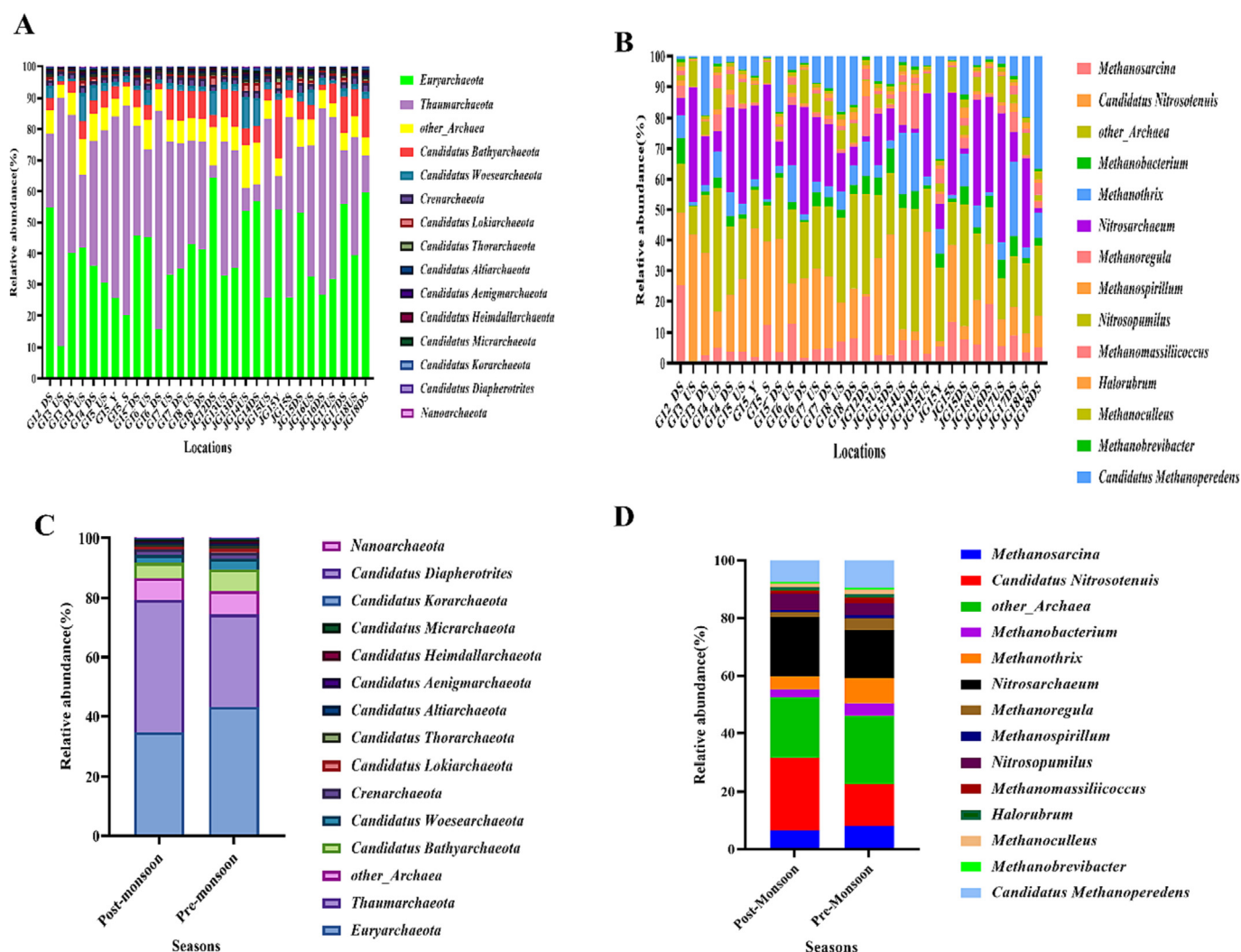


Fig. 5. Archaeal community composition across the lower stretch of River Ganges. 5A: Relative abundance of top 15 Archaeal phyla. 5B: Relative abundance of top 15 Archaeal genera. 4C, D: A seasonal comparison of the top 15 Archaeal phyla (5C) and top 15 Archaeal genera (5D) showing compositional homogeneity.

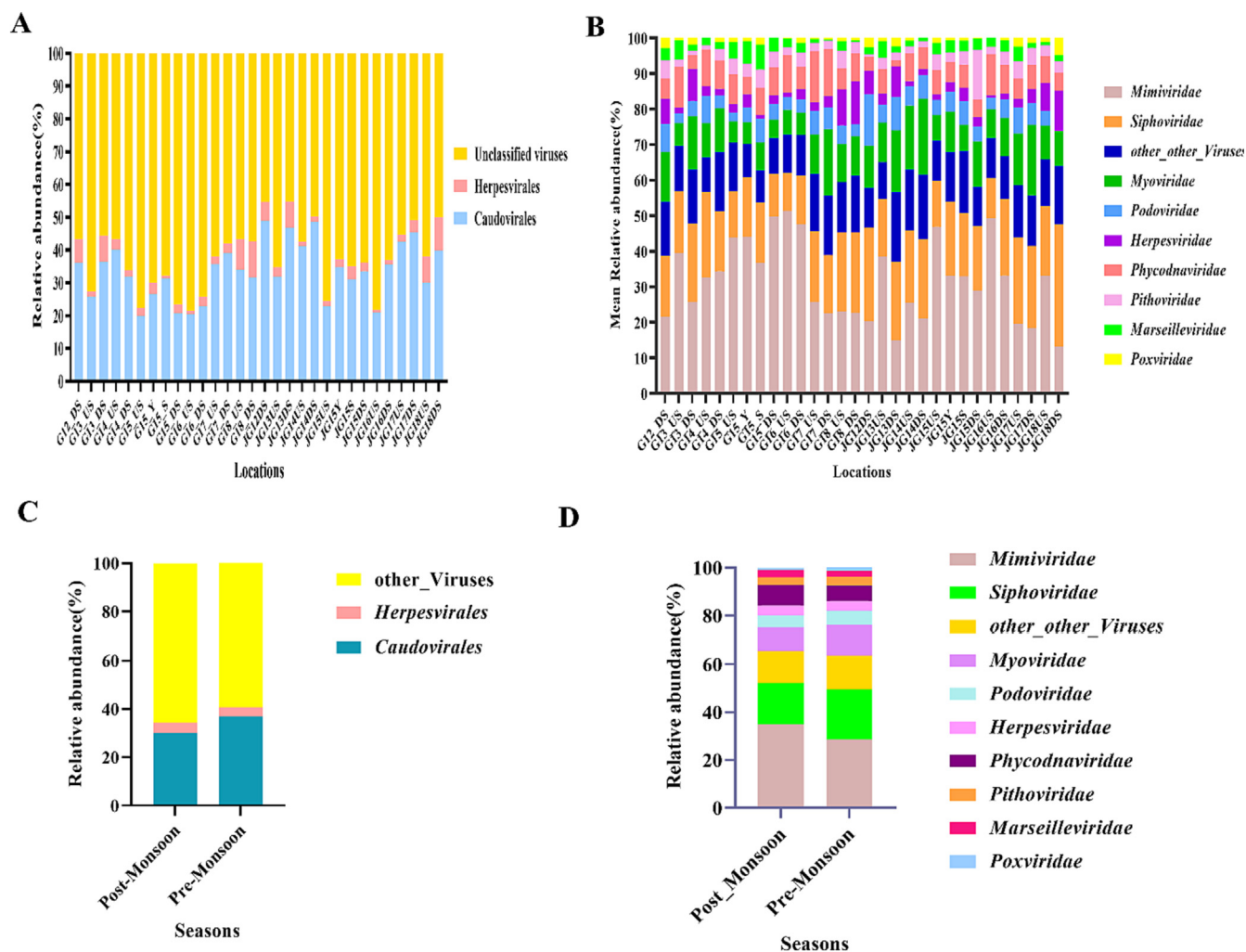


Fig. 6. Viral community composition across the lower stretch of the River Ganges. **6A:** Relative abundance of viral order. **6B:** Relative abundance of top 10 viral order. **6C, D:** A seasonal comparison of viral order (**6C**) and top 10 viral families (**6D**) showing compositional homogeneity.

to the ARG repertoire of the Ganges river at Varanasi (Reddy and Dubey, 2019). The diversity of the ARGs associated with the top 10 taxa has been tabulated in supporting information (Table S4). Besides, most of the *Burkholderia* and *Stenotrophomonas* species are important nosocomial pathogens and are frequently encountered with *Cupriavidus*, *Pandoraea*, and *Ralstonia* species from the respiratory specimens of cystic fibrosis patients (Lipuma et al., 2015). Various anthropogenic influences in the forms of hospital-associated discharges, faecal contamination, and mass bathing could be linked to the prevalence of these taxa. Therefore, these findings value consideration toward water quality monitoring and management of the River Ganges across the urban settlements.

MGEs are the drivers for the evolution and proliferation of multiple antibiotic-resistant bacteria (Chakraborty et al., 2020a). A maximum MGE diversity was observed in G13DS (110,000), G18US (125,000), and G18DS (150,000) as compared to other samples (Fig. S6A). This indicates profuse anthropogenic inputs leading to a higher horizontal gene transfer at these sites. Transposase (70.87 ± 5.80 %) was the most dominant MGE detected in the samples, followed by recombinase (15.28 ± 3.04 %) and integrase (7.65 ± 1.14 %) (Fig. S6B). Transposase enzymes from transposons are involved in the transmission of antibiotic resistance, which benefits the host's fitness in terms of survival, species diversification, and niche expansion. However, when human pathogens acquire such genes, it stakes public health (Khedkar et al., 2022).

3.6. Bacteriophage diversity and functions

River Ganges has been long known for its self-cleansing and non-putrefying properties, attributed primarily to its bacteriophages (phages) (Khairnar, 2016b). However, this river remains under-explored for the phage diversity across its anthropogenically impacted stretch. Targeting this facet, we annotated our data using MG-RAST to unveil the abundance and associated functions of the bacteriophage diversity. Analysis revealed the presence of bacteriophages from *Siphoviridae*, *Myoviridae* and *Podoviridae* families against antibiotic and metal-resistant bacteria such as *Pseudomonas*, *Burkholderia*, *Stenotrophomonas*, *Cupriavidus*, *Pandoraea*, and *Ralstonia*. As each location showed a substantial phage diversity, Krona plots in MG-RAST were used to comprehend their prevalence independently (Supporting file phage diversity).

Further, we grouped the host-phage data based on the occurrence of phages against putrefying bacteria, human pathogens and plant pathogens (Table S5). Based on this information, River Ganges can be explored to isolate bacteriophages against various human and plant pathogens. Also, phages and/or their enzymes obtained from this river can be used for the environmental remediation of other polluted rivers. Therefore, these findings are essential for mitigating various water-management and conservation policies for restoring the Ganges River and other polluted rivers. Furthermore, to understand the phage-related and host-page associated functions, predictive *in-silico*

functional annotation of the datasets was carried out using Subsystems and KEGG databases in MG-RAST (Fig. 8A). Prevalence of hits against phage-related proteins indicates a strong presence of a host-phage relationship. Interestingly, our study also found hits for sequences related to phage-associated lytic activities (lysozyme, endolysin, *N*-acetylmuramoyl-L-alanine amidase, endopeptidases) (Fig. 8B). These enzymes can be further used for bioprospection in therapy, diagnostics, and biotechnology related applications. As this is a preliminary study portraying phage diversity and its associated functions, further research is warranted to investigate and characterize the biological and genetic potentials of the phages for diverse applications.

4. Conclusions

- A higher prevalence of ARGs and MRGs suggested prolonged exposure of the microbial communities to anthropogenic pollutants, posing a severe threat to the riverine ecosystem.
- Important nosocomial relevant pathogens, like *Pseudomonas*, *Burkholderia*, and *Stenotrophomonas*, were the main potential taxa of ARGs and MRGs repertoire.
- A rich bacteriophage diversity at all locations could be further explored for their bioprospection toward the non-putrefying property of the River Ganges.

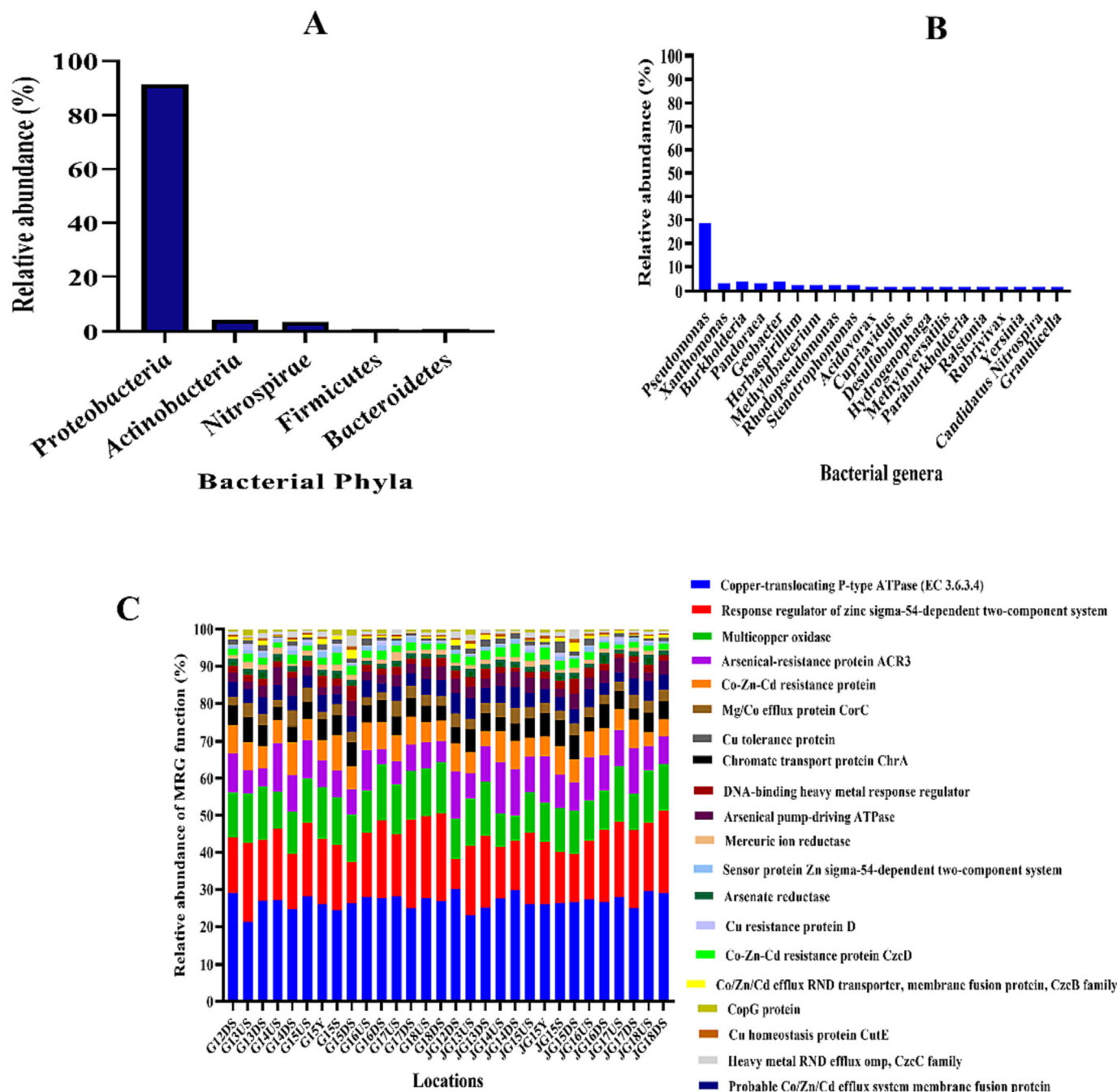


Fig. 7. Microbial diversity of taxa with metal resistance genes and the associated functions across the lower stretch of the River Ganges. 7A: Relative abundance of bacterial phyla conferring MRGs. 7B: Bacterial genera responsible for the MRG repertoire. 7C: Spatial composition of *in-silico* predictive functions associated with each MRG type.

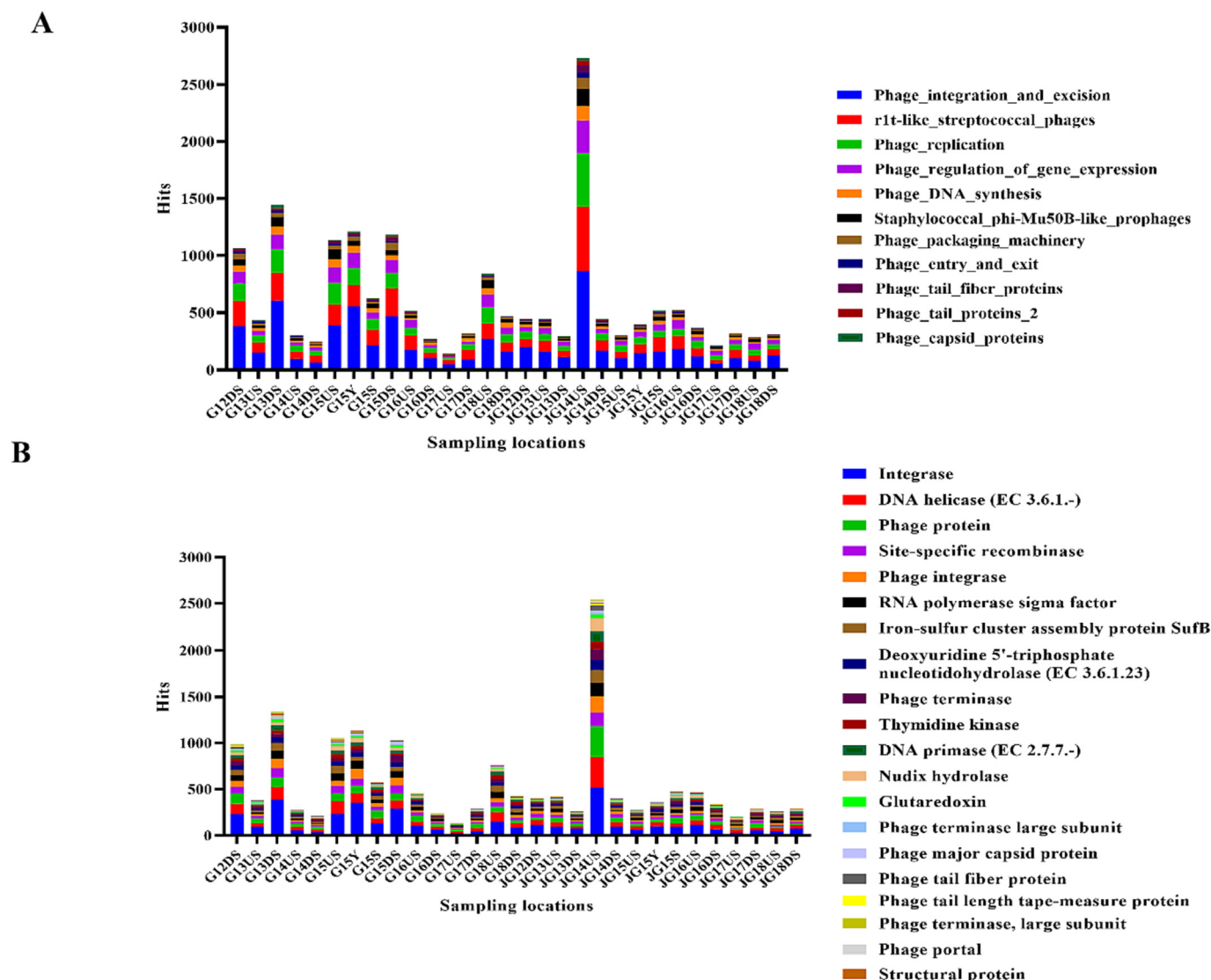


Fig. 8. Bacteriophage diversity and functions across the lower stretch of the River Ganges. **8A:** Predictive *in-silico* functional annotation of hits against host-phage-related proteins using MG-RAST. **8B:** Predictive functions of phage-associated lytic cassettes.

- The current study will significantly aid in the surveillance/ risk-based assessment of the River Ganges relating to long-term exposure to ARGs and multi-metal pollution.
- This data could be used in mitigating strategies for water pollution and improving the downstream trophic status of the River Ganges.

CRediT authorship contribution statement

KK, MSD, SGD: Conceptualization, Supervision, review and editing. **RS:** Visualization, investigation, sampling, initial data analysis, writing- review & editing; **VR:** Data analysis, review; **RKY:** Sampling, review **MS:** Sampling, review.

Data availability

The metagenomic datasets used in this study are publicly available in MG-RAST under the project name 'Ganga_Lower Stretchdata_NCL2019'. The MG-RAST accession IDs and sequence analysis stats for each dataset have been summarized (**Supporting file_seq stats**). This data has also

been submitted to Sequence Read Archive (SRA) portal under the bioproject PRJNA902095. The details of all the samples with their accession IDs have been collated in supporting information (**SRA biosample accessions_ PRJNA902095**).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2023.162125>.

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